



CAR SALES IN INDIA PRE-PANDEMIC (2019) AND POST-PANDEMIC(2021)



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Data Description:-

This dataset captures monthly car sales trends in India, covering the period before and after the onset of the COVID-19 pandemic. The goal is to analyze the impact of the pandemic on the automotive industry by comparing key indicators across these two phases.

The dataset typically includes data from January 2019 to December 2019 and January 2021 to December 2021 , split into:

- **Pre-Pandemic Period:** January 2019 – December 2019
- **After Pandemic Period:** January 2021 – December 2021

Data source:- Kaggle

<https://www.kaggle.com/datasets/subhadeeptasahoo/car-sales-in-india-20192021>

Data Type:-

It's Multivariate Monthly Time Series Data combining quantitative and categorical features.

OBJECTIVE:-

The objective is to take the monthly data of car sales and analyze it graphically and statistically ,and then take the quarterly data of that particularly dataset and analyze it in the same way.

Methodology:-

To get the result, we will apply chi-square test and Two-way ANOVA

DATA TO BE ANALYZE

MONTHLY DATA OF UNITS SALES OVER THE YEAR 2019 AND 2021

Company	Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Maruti Suzuki	2019	132872	129936	138483	129990	121012	111014	96478	93173	110454	139121	139133	122784
Hyundai	2019	45803	43110	44350	42005	42502	42007	39010	38205	40705	50010	44600	37953
Tata	2019	17272	17818	17714	12695	10900	13351	10485	7316	8097	13169	10400	12785
Mahindra	2019	22399	24520	25982	18901	19541	17762	16024	13147	13967	17923	14240	15276
Kia	2019	0	1	0	0	1	1	0	6236	7754	12854	14005	4645
Honda	2019	18259	13527	17201	11272	11442	10314	10250	8291	9301	10010	6459	8412
Toyota	2019	11221	11760	12818	10112	12138	10603	10423	10701	10203	11866	8312	6544
Renault	2019	5825	6241	7127	6256	5949	5400	3660	5704	8345	11516	10882	11964
Ford	2019	7700	6669	8271	6515	6361	5333	6284	5517	5556	7017	5392	3021
Nissan	2019	1500	774	977	518	476	401	313	288	295	401	350	617
MG	2019	1	0	0	1	0	0	1508	2018	2608	3536	3239	3021
Volkswagen	2019	2803	2617	2751	1995	2685	2463	2521	2306	2550	3213	2937	3483
Skoda	2019	1232	1267	1401	1126	1148	987	1076	1164	1233	1378	1266	2006
Jeep	2019	1267	1304	1441	1204	977	791	509	605	603	854	638	742

Company	Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Maruti Suzuki	2021	139002	144761	146203	135879	32903	124280	133732	103187	63111	108991	109722	123016
Hyundai	2021	52005	51600	52600	49002	25001	40496	48042	46866	33087	37021	37001	32312
Tata	2021	26980	27224	29655	25095	15181	24111	30184	28017	25729	33296	29780	35300
Mahindra	2021	20498	16702	16643	18285	8004	16636	20797	15786	12863	20034	19400	17476
Kia	2021	19056	15380	19100	16111	11050	15015	15016	16750	14441	16331	14214	7797
Honda	2021	11320	9324	7103	9072	2032	4767	6055	11177	6765	8108	5457	7973
Toyota	2021	11126	14069	14997	9621	707	8798	13103	12769	9284	12440	13002	10883
Renault	2021	8209	11043	12356	8642	2620	6100	9787	9703	7326	8910	5052	6130
Ford	2021	4141	5775	7746	5469	766	4936	3139	1508	0	0	1	0
Nissan	2021	4021	4244	4012	3369	1235	3336	4208	3170	2701	3619	2392	2783
MG	2021	3602	4329	5528	2565	1016	3558	4225	3900	3241	2863	2481	2550
Volkswagen	2021	2041	2186	1159	1533	1425	1633	1962	1631	2563	3077	3154	3700
Skoda	2021	1004	853	1159	961	716	734	3080	3829	3027	3065	2196	3233
Jeep	2021	394	1103	1360	846	475	789	909	1173	1311	1324	1052	916

QUARTERLY DATA OF UNITS SALES OVER THE YEAR 2019 AND 2021

Company	Year	Quarter 1	Quarter 2	Quarter 3	Quarter 4
Maruti Suzuki	2019	401291	362016	300105	401038
Hyundai	2019	133263	126514	117920	132563
Tata	2019	52804	36946	25898	36354
Mahindra	2019	72901	56204	43138	47439
Kia	2019	1	2	13990	31504
Honda	2019	48987	33028	27842	24881
Toyota	2019	35799	32853	31327	26722
Renault	2019	19193	17605	17709	34362
Ford	2019	22640	18209	17357	15430
Nissan	2019	3251	1395	896	1368
MG	2019	1	1	6134	9796
Volkswagen	2019	8171	7143	7377	9633
Skoda	2019	3900	3261	3473	4650
Jeep	2019	4012	2972	1717	2234

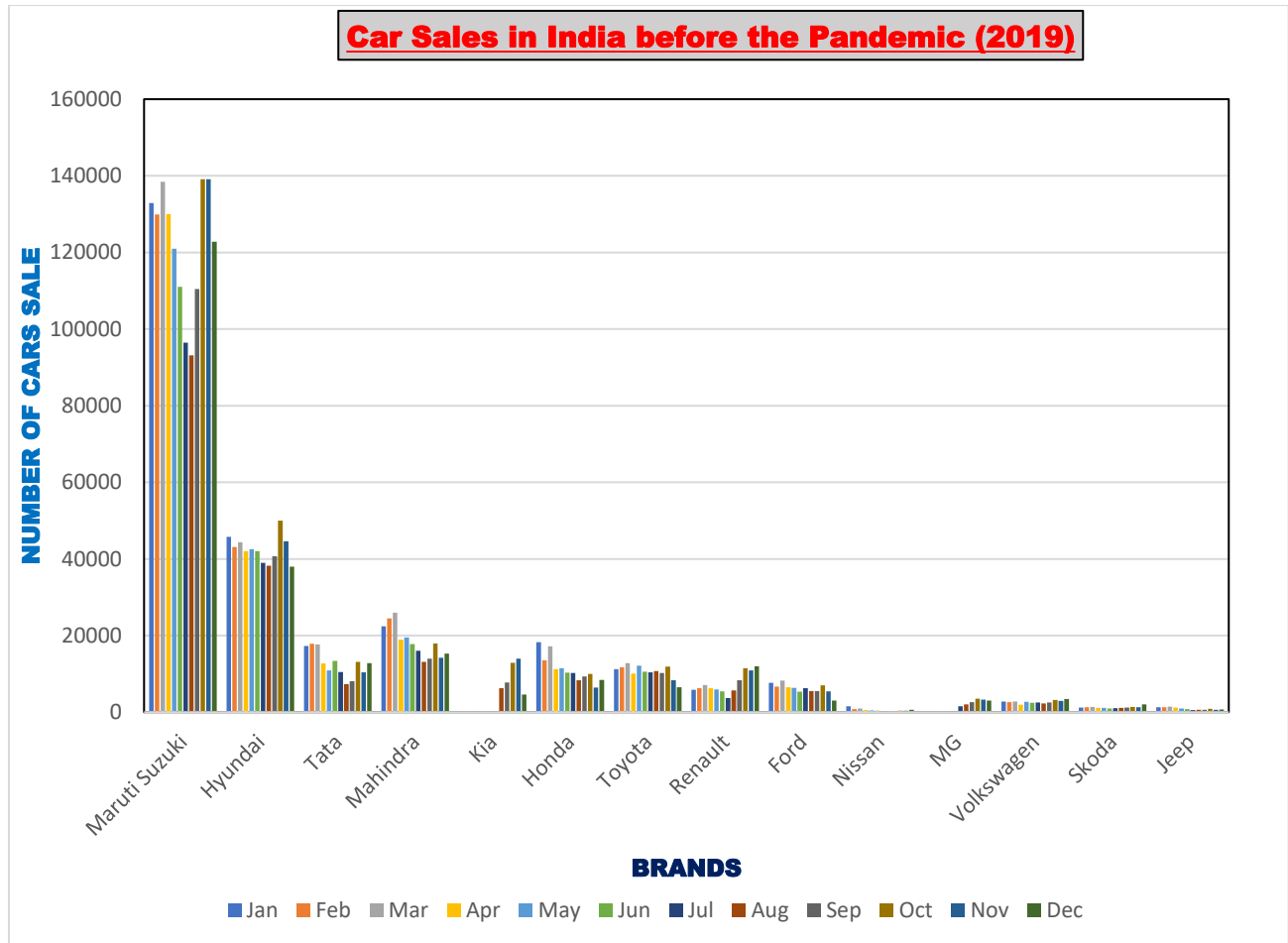
Company	YEAR	Quarter 1	Quarter 2	Quarter 3	Quarter 4
Maruti Suzuki	2021	429966	293062	300030	341729
Hyundai	2021	156205	114499	127995	106334
Tata	2021	83859	64387	83930	98376
Mahindra	2021	53843	42925	49446	56910
Kia	2021	53536	42176	46207	38342
Honda	2021	27747	15871	23997	21538
Toyota	2021	40192	19126	35156	36325
Renault	2021	31608	17362	26816	20092
Ford	2021	17662	11171	4647	1
Nissan	2021	12277	7940	10079	8794
MG	2021	13459	7139	11366	7894
Volkswagen	2021	5386	4591	6156	9931
Skoda	2021	3016	2411	9936	8494
Jeep	2021	2857	2110	3393	3292

Analysis of data

- Graphical Interpretation
- Statistical Analysis

Graphical Interpretation

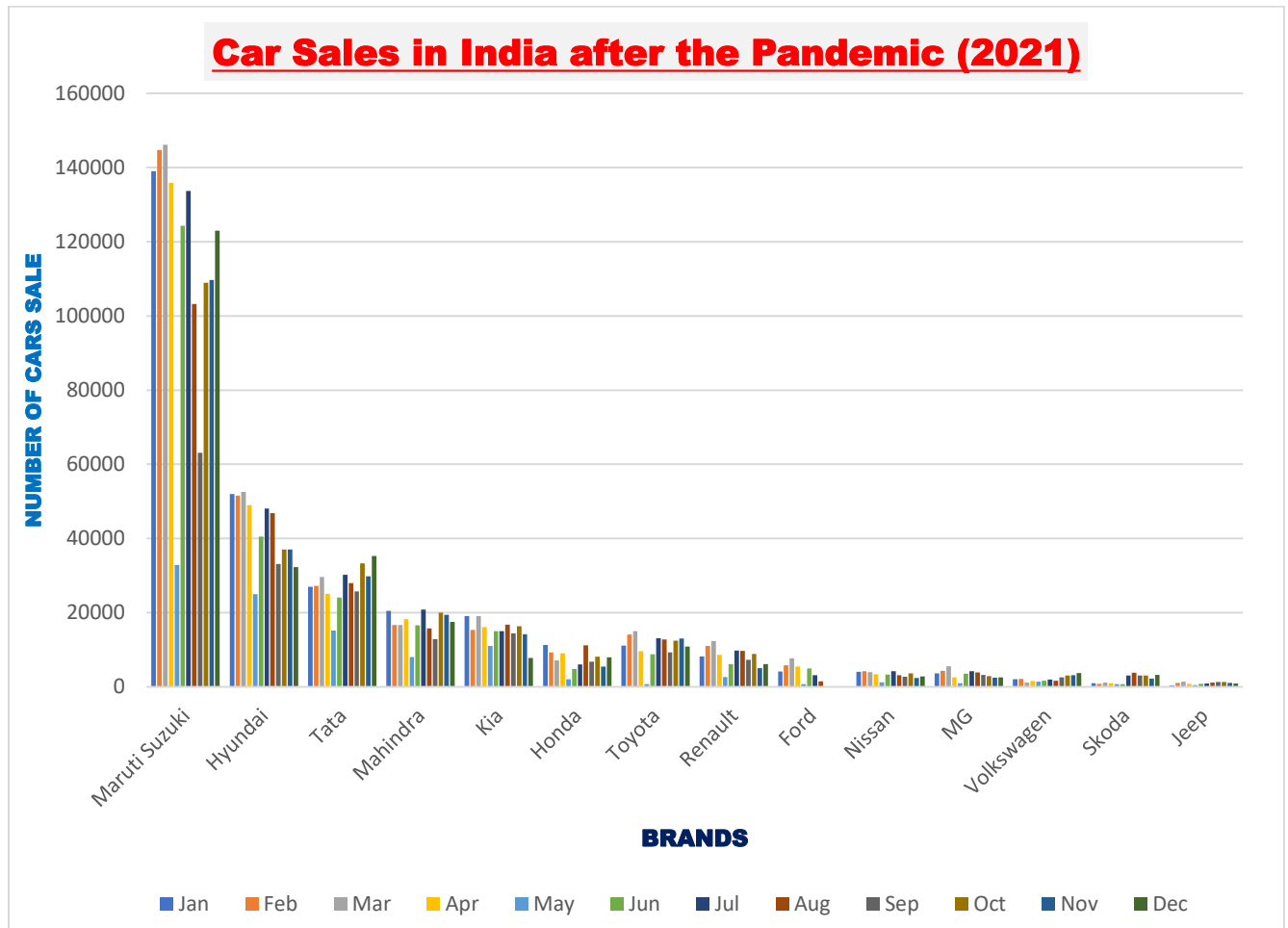
Plot of the monthly units sale over year (2019)



Interpretation:-

The bar chart shows monthly car sales in India for various brands during 2019. Maruti Suzuki and Hyundai clearly dominate the market with consistently high monthly sales, while brands like Jeep, Skoda, and Volkswagen have significantly lower sales throughout the year.

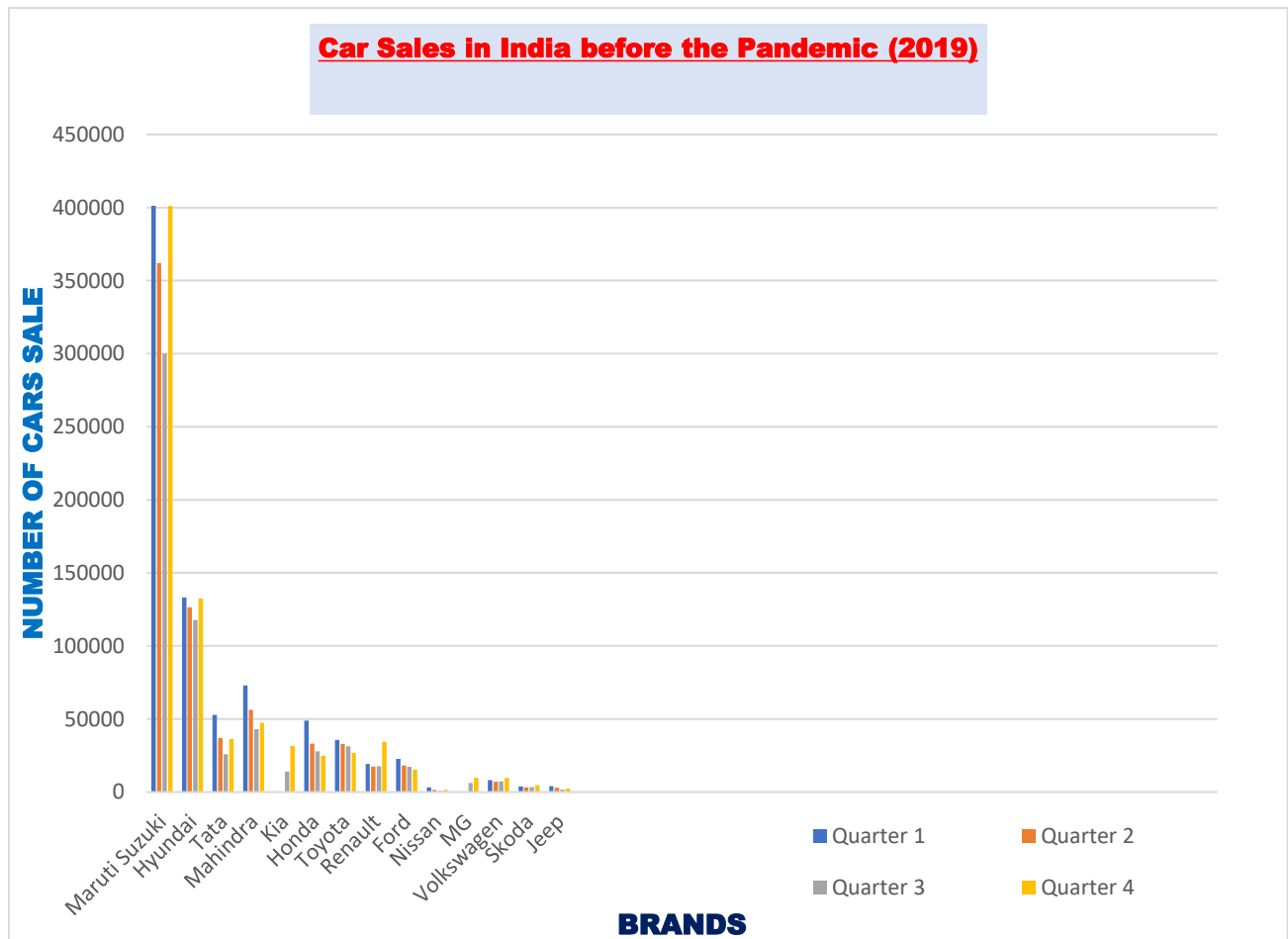
Plot of the monthly units sales over year (2021)



Interpretation:-

The bar graph shows that Maruti Suzuki had the highest car sales in India throughout 2021, followed by Hyundai, Tata, and Mahindra. Most other brands had significantly lower sales. This suggests that the Indian car market post-pandemic in 2021 was dominated by a few major players.

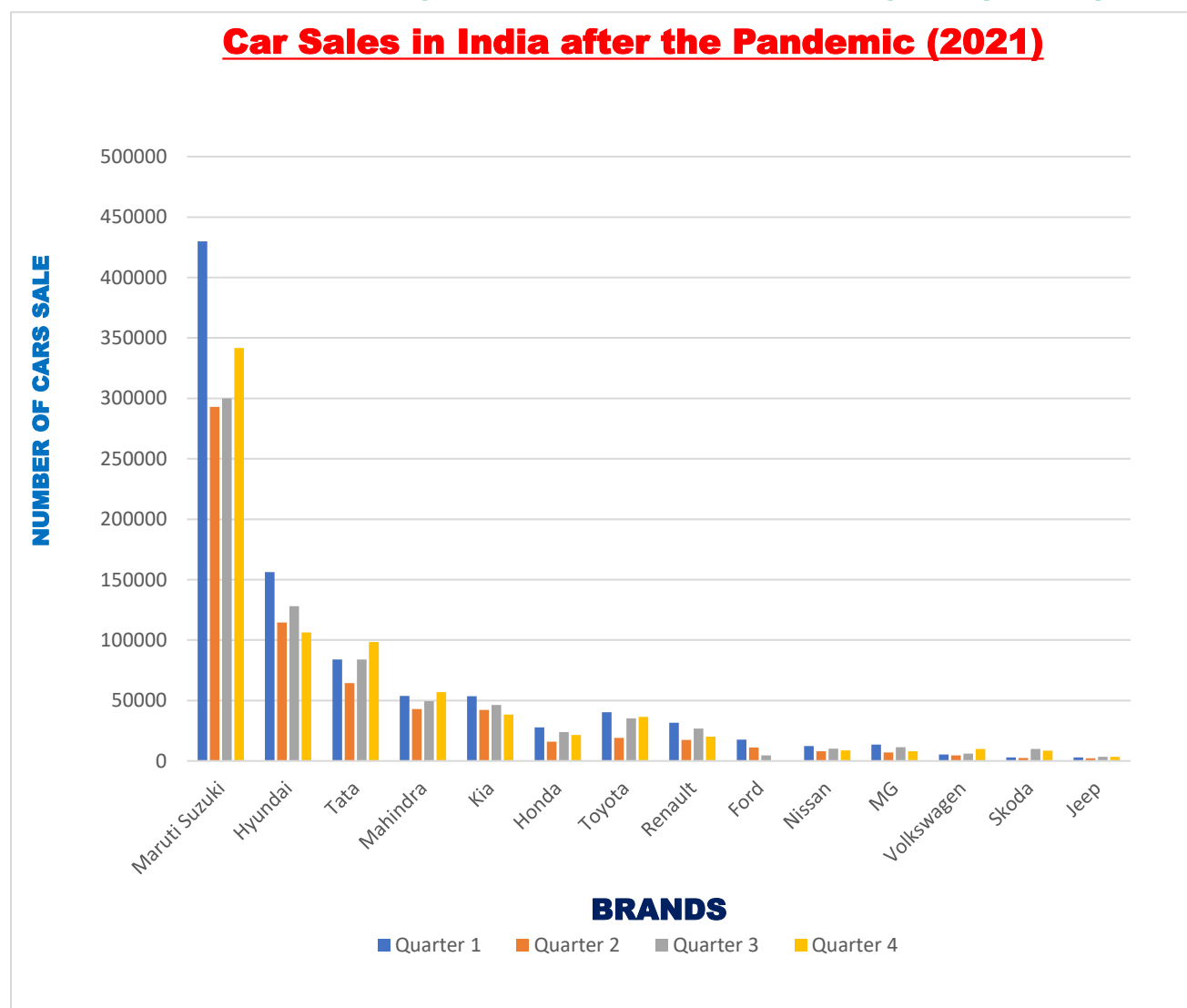
Plot of the quarterly units sales over the year(2019):-



Interpretation:-

The 2019 graph shows that Maruti Suzuki had significantly higher quarterly car sales than all other brands, such as Hyundai, Mahindra, and Tata. Overall sales volumes were much higher before the pandemic, indicating a stronger market in 2019 compared to post-pandemic 2021.

Plot of the quarterly units sales over the year(2021):-



Interpretation:-

The 2021 quarterly sales graph shows Maruti Suzuki remained the top-selling brand, with Hyundai, Tata, and Mahindra trailing behind. Compared to 2019, overall sales were lower, indicating a partial market recovery post-pandemic but still below pre-pandemic levels.

Statistical Analysis



Pearson Chi-Square Test on monthly car sales data for 2019 and 2021

Dataset for Pearson chi-square test

Company	Year(2019)	Year(2021)
Maruti suzuki	1464450	1364787
Hyundai	510260	505033
Tata	152002	330552
Mahindra	219682	203124
Kia	45497	180261
Honda	134768	89153
Toyota	126701	130799
Renault	88869	95878
Ford	73636	33481
Nissan	6910	39090
MG	15932	39858
Volkswagwen	32324	26064
Skoda	15284	23857
Jeep	10935	11652

To test whether the sales on the brand does depend on the year:-

We consider :-

H₀(Null hypothesis): Years and the brands of the cars are independent in terms of count of sale.

Against

H₁(Alternative hypothesis): Years and the brands of the cars are depends in terms of count of sale.

As we knpow that,

$$\text{Chi-square} = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where:

O_{ij}= Observed frequenc

E_{ij}= Expected Frequency

$$E_{ij} = \frac{(\text{Row total } i) * (\text{Column total } j)}{\text{Grand Total}}$$

Company	Observed2019(O1)	Expected 2019(E1)	Observed 2021(O2)	Expected 2021 (E2)
Maruti Suzuki	1464450	1371869	1364787	1457368
Hyundai	510260	492481	505033	522812
Tata	152002	234132	330552	248422
Mahindra	219682	205205	203124	217601
Kia	45497	109477	180261	116281
Honda	134738	108637	89153	115254
Toyota	126701	124940	130799	132560
Renault	88869	89674	95878	95073
Ford	73636	51989	34381	55128
Nissan	6910	22327	39090	23673
MG	15932	27076	39858	28714
Volkswagen	32324	28336	26064	30052
Skoda	15284	19002	23857	20139

Company	Observed2019(O1)	Expected 2019(E1)	Observed 2021(O2)	Expected 2021 (E2)
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Jeep	10935	10989	11652	11598
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Now we will calculate chi-square test statistic for each year to the corresponding brands

$$\text{Chi-square} = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$= 56017.97 + 58551.69 + 633.94 + 604.14 + 28779.04 + 27103.57 + 1044.38 + 989.20 + 38284.77 + 35873.27 + 6650.36 + 5639.17 + 24.92 + 22.26 + 6.45 + 6.80 + 17416.21 + 7730.95 + 11265.55 + 9989.90 + 4597.84 + 4576.43 + 560.18 + 527.58 + 728.13 + 701.41 + 0.27 + 0.26$$

Total Chi-square statistic ≈ 377011.70

Degrees of Freedom

Formula:

$$df = (r - 1)(c - 1)$$

where:

- $r = 14$ (number of brands)
- $c = 2$ (years: 2019 and 2021)

Thus:

$$df = (14 - 1)(2 - 1) = 13$$

Conclusion:-

Now we have the critical value of chi square at 13 degrees of freedom is 22.362 and the chi square value = 377011.70

since,

chi-square calculated = 377011.70 >> 22.362

or p-value < 2.2e-16 < 0.05

We reject the null hypothesis

This implies that Years and the brandwise sales are not independent rather very low p-value states over the year brand specific sales was changed

for example,

there was craze for ford car in 2019 while 2021 that craze had gone off.

Pearson Chi-Square Test on Quarterly car sales data for 2019 :-

Datasets for 2019:

<u>Company (Brand)</u>	<u>Quarter 1</u>	<u>Quarter 2</u>	<u>Quarter 3</u>	<u>Quarter 4</u>
Maruti suzuki	401291	362016	300105	401038
Hyundai	133263	126514	117920	132563
Tata	52804	36946	25898	36354
Mahindra	72901	56204	43138	47439
Kia	1	2	13990	31504
Honda	48987	33028	27842	24881
Toyota	35799	32853	31327	26722
Renault	19193	17605	17709	34362
Ford	22640	18209	17709	34362
Nissan	3251	1395	896	1368
MG	1	1	6134	9796
Volkswagwen	8171	7143	7377	9633
Skoda	3900	3261	3473	4650
Jeep	4012	2972	1717	2234

Now we will test the Quarterwise sales reports for different brands of cars

Let,

H_0 (null hypothesis):- Whether the quarterwise sales is independent of the brands of the cars

H_1 (Alternative hypothesis):- The Quarterwise sales depends on the brands of the cars

Pearson's Chi-squared test

data: table

X-squared = 105458, df = 39, p-value < 2.2e-16

since :

we have

p-value < 2.2e-16 < 0.05

We reject the null hypothesis,

Quarterwise sales and brands of the cars are not independent.

Pearson Chi-Square Test on Quarterly car sales data for 2021:-

Datasets for 2021:

<u>Company (Brand)</u>	<u>Quarter 1</u>	<u>Quarter 2</u>	<u>Quarter 3</u>	<u>Quarter 4</u>
Maruti suzuki	429966	293062	300030	341729
Hyundai	156205	114499	127995	106334
Tata	83859	64387	83930	98376
Mahindra	53843	42925	49446	56910
Kia	53536	42176	46207	38342
Honda	27747	15871	23997	21538
Toyota	40192	19126	35156	36325
Renault	31608	17362	26816	20092
Ford	17662	11171	4647	1
Nissan	12277	7940	10079	8794

MG	13459	7139	11366	7894
Volkswagwen	5386	4591	6156	9931
Skoda	3016	2411	9936	8494
Jeep	2857	2110	3393	3292

Let,

H₀(null hypothesis):- Whether the quarterwise sales is independent of the brands of the cars

H₁(Alternative hypothesis):- The Quarterwise sales depends on the brands of the cars

Pearson's Chi-squared test

data: table

X-squared = 53408, df = 39, p-value < 2.2e-16

since :

we have

p-value < 2.2e-16 < 0.05

We reject the null hypothesis,

Quarterwise sales and brands of the cars are not independent. rather very low p-value over the quarter.

- **Levene's test:**

- Levene's Test is the statistical test which is used to test the assess the equality of variances (Homoscedasticity) across two or more groups.
- If the P- value of levene's test is less than the chosen significance level (Here we take alpha= 0.05), it suggest that the variances across group are significantly different.
- We get from R,
- Levene's Test for quarters sales for 2019 and 2021

Levene's Test for Quarters:

```
> leveneTest(carsale ~ Company, data = df)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	13	0	1
	182		

Levene's Test for Quarters:

```
> leveneTest(carsale ~ Company, data = df)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	13	0	1
	182		

```
>
```

DATA FOR TWO WAY ANOVA (2019):-

For 2019 we analyze the variation

- (i) Between Quarters(Seasonal)
- (ii) Between Car Brands(Company-wise variation)
- (iii) And intraction of both

Data Structure

<u>Company (Brand)</u>	<u>Quarter 1</u>	<u>Quarter 2</u>	<u>Quarter 3</u>	<u>Quarter 4</u>	<u>Row means</u> $\bar{y}_{i0}$
Maruti suzuki	401291	362016	300105	401038	366112.50
Hyundai	133263	126514	117920	132563	127565.00
Tata	52804	36946	25898	36354	38000.50
Mahindra	72901	56204	43138	47439	54920.5
Kia	1	2	13990	31504	11374.25
Honda	48987	33028	27842	24881	33684.5
Toyota	35799	32853	31327	26722	31675.25
Renault	19193	17605	17709	34362	22217.25
Ford	22640	18209	17709	34362	18409
Nissan	3251	1395	896	1368	1727.50
MG	1	1	6134	9796	3983
Volkswagwen	8171	7143	7377	9633	8081
Skoda	3900	3261	3473	4650	3821
Jeep	4012	2972	1717	2234	2733.75
Column means ($\bar{y}_{0j}$)	57586.71	49867.785	43920.214	55569.57	$\bar{y}_{00}=51540.80357$

- 14 car brands = r

- 4 quarters = c
- Total observations = 14 car brands * 4 Quarters = 56

Let consider the Testing Hypothesis

For Companies (brand effect):-

H_0 (Null hypothesis): No significant difference in sales across companies

H_1 (Alternative hypothesis): At least one significant brand differs significantly

For Quarters (seasonality)

H_0 (Null hypothesis): No significant difference in sales across quarters

H_1 (Alternative hypothesis): Atleast one quarter differs significantly

We have ,

Y_{ij} = sales of the cars of i^{th} brand in j^{th} quarter

$\bar{y}_{00}$ = Grand mean

$\bar{y}_{i0}$ = row mean(brand mean)

$\bar{y}_{0j}$ = column mean(quarter mean)

Now,

Total sum of square (SST) = $\sum_{i=1}^r \sum_{j=1}^c (y_{ij} - \bar{y}_{00})^2 = 491,239,314,387.43$

Sum of square for Brands(SSB) = $c \sum_{i=1}^r (\bar{y}_{i0} - \bar{y}_{00})^2 = 482,000,453,501.43$

Sum of square for quarters (SSQ) = $r \sum_{j=1}^4 (\bar{y}_{0j} - \bar{y}_{00})^2 = 1,582,229,777.29$

Error sum of square(SSE) = SST-SSB-SSQ = 7,656,631,108.71

ANOVA TABLE:-

Source of variation	d.f	SS	MS	F _{obs}	F _{tab}
Brands	(r-1)=14-1=13	SSB= 482,000,453,501.43	$MSB = \frac{SSB}{13} =$ 37076957961.65	$\frac{MSB}{MSE} =$ 188.86	F _{0.05,13,39} = 2.34
Quarters	(c-1)=4-1=3	SSQ= 1,582,229,777.29	$MSQ = \frac{SSQ}{3} =$ 527409925.76	$\frac{MSQ}{MSE} =$ 2.69	F _{0.05,13,3} = 3.14
Error	(r-1)*(c-1) =39	SSE= 7,656,631,108.71	$MSE = \frac{SSE}{39} =$ 196323874.58		
Total	55	SST= 491,239,314,387.43			

As we get,

For brands:-

$$F_{obs} > F_{tab} \Rightarrow 188.86 > 2.34$$

So we reject the null hypothesis

For Quarters:-

$$F_{obs} < F_{tab} \Rightarrow 2.69 < 3.14$$

So we fail to reject the null hypothesis

There is significant no significant difference on car sales across quarters.

DATA FOR TWO WAY ANOVA (2021):-

similary for 2021 we wil analyze the variation

Data Structure

Company (Brand)	Quarter 1	Quarter 2	Quarter 3	Quarter 4	Row means $\bar{Y}_{i0}$
Maruti suzuki	429966	293062	300030	341729	341196.75
Hyundai	156205	114499	127995	106334	126258.25
Tata	83859	64387	83930	98376	82638
Mahindra	53843	42925	49446	56910	50781
Kia	53536	42176	46207	38342	45065.25
Honda	27747	15871	23997	21538	22288.25
Toyota	40192	19126	35156	36325	32699.75
Renault	31608	17362	26816	20092	23969.5
Ford	17662	11171	4647	1	8370.25
Nissan	12277	7940	10079	8794	9772.50
MG	13459	7139	11366	7894	9964.5
Volkswagwen	5386	4591	6156	9931	6516
Skoda	3016	2411	9936	8494	5964.25
Jeep	2857	2110	3393	3292	2913
Column means $\bar{Y}_{0j}$	66543.78571	46055	52796.71429	54146.57143	$\bar{Y}_{00} =$ 54885.51786

- 14 car brands = r
- 4 quarters =c
- Total observations = 14 car brands * 4 Quraters = 56

Let consider The Testing of Hypothesis

For Quarters (seasonality)

H_0 (Null hypothesis): No significant difference in sales across quarters

H_1 (Alternative hypothesis): Atleast one quarter differs significantly

For Companies (brand effect):-

H_0 (Null hypothesis): No significant difference in sales across companies

H_1 (Alternative hypothesis): At least one significant brand differs significantly

Now,

Total sum of square (SST)= $\sum_{i=1}^r \sum_{j=1}^c (y_{ij} - \bar{y}_{00})^2 = 431335476000$

Sum of square for Brands(SSB) = $c\sum_{i=1}^r (\bar{y}_{i0} - \bar{y}_{00})^2 = 416452238200$

Sum of square for quarters (SSQ) = $r\sum_{j=1}^4 (\bar{y}_{0j} - \bar{y}_{00})^2 = 3063234668$

Error sum of square(SSE) = SST-SSB-SSQ = 11820003130

ANOVA Table:-

Source of variation	d.f	SS	MS	F_{obs}	F_{tab}
Brands	$(r-1)=14-1=13$	SSB=416452238200	$MSB=\frac{SSB}{13}=32034787550$	$\frac{MSB}{MSE}=105.70$	$F_{0.05,13,39} = 2.34$
Quarters	$(c-1)=4-1=3$	SSQ=3063234668	$MSQ=\frac{SSQ}{3}=1021078228$	$\frac{MSQ}{MSE}=3.37$	$F_{0.05,13,3} = 3.14$
Error	$(r-1)*(c-1)=39$	SSE=11820003130	$MSE=\frac{SSE}{39}=303077003.3$		
Total	55	SST=431335476000			

Now we get,

For brands:-

$F_{obs} > F_{tab}$

So we reject the H_0

It means that at least one significant brand differs significantly

For Quarters:-

$$F_{obs} > F_{tab}$$

So we reject the H_0

There is a significant difference in sales across brands and quarters.

CONCLUSION:-

On the basis of 2019 and 2021 ANOVA summary we can check whether the pandemic had a significant impact.

- **Brand Effect:-**
 - As per our analysis we can say that there is a significant differences in the sales
 - of the different brands of cars in 2019 and 2021 ,rather it can be said that Maruti and Hyundai dominated to the sales of remain brands of cars.
- **Quarter Effect:-**
 - Not significant in the year 2019 but in 2021. it means that sales do not significantly fluctuate by quarters in 2019 but fluctuate in 2021, even during the pandemic.
- **Impact of Pandemic:-**
 - Overall sales figure dropped significantly in 2019 and 2021 for most brand (e.g, Hyundai, Honda Ford.)
 - Entry-level brands like Kia and MG grow during 2021.
 - While the pattern of variation in sales remained the same overall sales volumes declined, reflecting the true pandemic impact.

REFERENCE:-

Fundamental of Statistics Volume One and two

- A.M Gun
- M.K Gupta
- B.Dasgupta

APPENDIX:-

```
1 # sales on the brand does depends on year
2 data.frame<-read.csv("C:/Users/Shivam Rawat/OneDrive/Desktop/6th SEM/car-sales-india-Original2019ans2019data for test.csv",header=TRUE)
3 data.frame
4 table<-as.matrix(data.frame[,2:3])
5 rownames(table)<-data.frame$COMPANY
6 print(table)
7 chi_result<-chisq.test(table)
8 print(chi_result)
9
10 #Quarterwise Sales reports(2019)
11 data.frame<-read.csv("C:/Users/Shivam Rawat/OneDrive/Desktop/6th SEM/car-sales-india-Original2019ans2019data for test.csv",header=TRUE)
12 data.frame
13 table<-as.matrix(data.frame[,2:5])
14 rownames(table)<-data.frame$COMPANY
15 print(table)
16 chi_result<-chisq.test(table)
17 print(chi_result)
18
19
20 #Quarterwise Sales reports(2021)
21 data.frame<-read.csv("C:/Users/Shivam Rawat/OneDrive/Desktop/6th SEM/car-sales-india-Original Quarterwisedata2021test.csv",header=TRUE)
22 data.frame
23 table<-as.matrix(data.frame[,2:5])
24 rownames(table)<-data.frame$COMPANY
25 print(table)
26 chi_result<-chisq.test(table)
27 print(chi_result)
28
29
30
```

```
1 #Levene's test for homoscedastity for Quarterly data of 2019
2 #Data
3 carsale<-c(1464450,510260,152002,219682,45497,134738,126701,88869,73636,6910,15932,32324,15284,10935)
4
5
6 Company <- factor(rep(c("Maruti Suzuki","Hyundai","Tata","Mahindra","Kia","Honda","Toyota",
7 "Renault","Ford","Nissan","MG","Volkswagen","Skoda","Jeep"),each = 14))# for 2019
8
9 df <-data.frame(carsale,Company )
10
11 #Levene's Test for equality of variances across different quarters
12 cat("Levene's Test for Quarters:\n")
13 leveneTest(carsale ~ Company, data = df)
14
15
16 #Levene's test for homoscedastity for Quarterly data of 2021
17 #Data
18 carsale<-c(1364787, 505033, 330552, 203124, 180261, 89153,130799, 95878, 33481, 39090,39858,26064, 23857,11652)
19
20 Company <- factor(rep(c("Maruti Suzuki","Hyundai","Tata","Mahindra","Kia","Honda","Toyota",
21 "Renault","Ford","Nissan","MG","Volkswagen","Skoda","Jeep"),each = 14))# for 2021
22
23 df <-data.frame(carsale,Company )
24
25 #Levene's Test for equality of variances across different quarters
26 cat("Levene's Test for Quarters:\n")
27 leveneTest(carsale ~ Company, data = df)
28
```